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LSTM-Based Hourly Sales Forecasting in SME Food and Beverage Retail

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***Abstract*—Another important aspect of inventory management in food and beverage business based on retailing is accurate demand forecasting. This paper will attempt to forecast hourly ice cream products sales in an SME using the Long Short-Term Memory (LSTM) model, and to compare the performance of the Long Short-Term Memory with cross-sectional approaches, i.e., Naive Forecast, 7-hour Moving Average (MA7), and Daily Seasonal Naive. The data on the transactions are processed by the Point-of-Sale (POS) system and arranged into hourly time units to record the daily demand dynamics. The preliminary examination shows that there is a robust seasonal trend of the day and that there is a high variation in sales during the working hours. On the test set, LSTM attains the best performance with MAE and RMSE, which is smaller than Naive and MA7. The LSTM model is also demonstrated to have the capability of reducing possible losses as a result of over or understocking compared to the two approaches through cost-based simulations.**

***Index Terms*—LSTM, time series forecasting, hourly sales prediction, inventory optimization**

I. Introduction

The level of competition in the business environment is getting more and more intense, especially within the food and beverage industry. There are numerous huge franchise brands with massive networks and well-established marketing strategies that rule the market. Small and medium enterprises in such a scenario are left to handle their operations effectively so that they can sustain and expand their businesses. An effective operational management involves proper analysis of the inventory, use of right raw material in order to avoid wastage and planning of sales as a result of demand analysis in the market and consumer trends. The strategies are likely to boost the competitiveness of smaller businesses, even where they have to compete directly with other players in the culinary industry [1].

The problem of poor demand forecasting can be manifested in either overstock or understock situation to the management of the inventory. This is a major concern among the tiny and medium businesses. Excessive stocking leads to the rise of storage expenses and the spoilage or degradation of the product. On the contrary, stockouts minimize customer satisfaction and create missed sales opportunities [2].

The correct predictive models make businesses accurately predict and estimate the daily, weekly and monthly product demand. This facilitates better buying decisions of the raw materials and minimize chances of overstocking or understocking. Adjustment of production activities can be also optimized more accurately that helps in optimization of the operational processes and provide quick response to changes in the market. [3].

Long Short-Term Memory or Random Forest are technologies which facilitate further analysis through historical data to draw more trustworthy predictions. By utilizing these strategies, the small to medium enterprises will be able to expand their knowledge on production planning and inventory management, thus increasing efficiency in their operations and competitive advantage.

This research article is centered on one of the culinary industry firms dealing with selling ice cream as a primary product. The goal is to offer real-life assistance in the optimization of inventory management using data-based approach. Based on the available sales data, this research will compare methods of prediction, Long Short-Term Memory and other base prediction technique to identify the technique that is more effective and accurate in predicting inventory demand in small-scale enterprises and enhancing operational efficiency and stock management.

The objective of this study is to create and test an LSTM-based demand forecasting model using hourly sales data, to compare it with conventional baseline methods, and to consider the potential benefits of the business in reducing under-stock and over-stock costs in stock decision making and sales strategies.

This study also examines the limitations of simple forecasting methods such as Naive Forecast and Moving Average, which are still frequently used in MSMEs. This occurs because these two methods are less able to capture strong daily seasonal patterns in hourly sales data. Therefore, the researchers compared the LSTM model with several of these baselines to demonstrate its practical contribution to assisting inventory decision-making at the operational level of businesses.

II. Related Work

Long Short-Term Memory networks are recurrent neural networks that overcome the weaknesses of the traditional recurrent systems, namely, the vanishing and exploding gradient issues that constrain the property of recurrent models of simulating long-range connectivity of sequential information. The networks make use of a unique structure that incorporates nodes and a gating mechanism that regulates a flow of information into, within or out of the cell. This architecture makes Long Short-term Memory networks to shine in problems involving modeling of temporal sequence like speech recognition, handwriting recognition and natural language processing tasks [4]. The model capacity is also improved by the bidirectional and stacked extensions as well as the contextual information contributed by the architectural extensions of the model [5]. These benefits notwithstanding, Long Short-Term Memory networks are computationally expensive and this is a problem with the implementation on resource-constrained systems. Efforts to alleviate this interest in doing away with redundant computation and hardware accelerator to enhance efficiency. [6].

Some studies have applied Long Short-Term Memory networks to forecast sales, demand of products or inventory of groceries and have shown that the networks considerably enhance the accuracy of the prediction in various retailing environments. These models are useful in modeling time dependent relationships and nonlinear trends that are common with sales data and can be easily defeated by traditional statistical methods. As illustrated by Agrahari [7], Long Short-Term Memory architectures have an advantage over linear regressions and decision tree models in that they are better-adjusted to time-series behavior, leading to better sales forecasting and more responsive business decisions.

In another study, Subhalakshmi et al. [8] developed a Long Short-Term Memory based prediction model of supermarkets, which was far more successful as it acknowledged variables such as store type, date, and item level sales and therefore it was more successful in inventory management and it was in a position to eliminate overstocking or stockouts losses. In the e-grocery domain, Golabek et al. [**?**] discovered that Long Short-Term Memory models were more accurate compared to machine learning and statistical models, but random forest and linear regression were slightly more accurate on beverages only. The implication of these findings is that the product-level demand forecasting with the help of Long Short-Term Memory networks offers a valid approach of inventory control and customer demand fulfillment.

One of the most common methods is naive forecasting [9] in which the value forecasted in the next period would be identical to the value that has been observed in the previous period. This method, no matter how simple it is, can sometimes work quite well with data with strong short-term correlations, such as the hourly loads in food and drinks retail [10]. In addition to this method, moving averages are also typically used as a baseline to an extent of smoothing out short term variation and also identify local trends [11]. The event of interest in the study was the sales average of the last seven hours (MA7) because this is the time period of the demand dynamics of the cafe within one daily cycle of activity. However, the approach is less efficient in reflecting recurrent routine seasonal patterns.

To correct this deficiency, the daily seasonal naive method [12] was used as a comparison also. It is a forecasting method which predicts demand using the value at the same hour of the day before, and it is also particularly relevant where the data has a strong daily seasonal cycle. This method suits well to hourly Point-of-Sale (POS) data, as customer demand patterns tend to stay constant on a day to day basis. Having the three baselines of naive forecast, MA7 and daily seasonal naive in place, the study provides further argument that the increased performance of the LSTM model could not be explained by the complexity of the model, but by the ability to capture more complex patterns in the product demand data, both time and season.

III. Methodology

In common time series forecasting practice, the selection of a modeling strategy typically begins with a comparison of a data-driven learning model with a statistical baseline method. This is done to ensure that the achieved performance improvements can be more clearly identified by the model's ability to learn the temporal patterns and structure of the data, thus facilitating interpretation of the effectiveness of the method used.

This paper involves sales transaction information of the Point-of-Sale (POS) system in one of the small business of ice drinks shops in Indonesia. The information will include all the transactions in the period 2023-2024, which will be in the form of monthly Excel files. The data sheets included in each file are called Transactions_Items and carry the following details like the transaction code, the time of transaction (timestamp), name of the product, quantity sold, selling price and method of payment. Among the products sold, this paper has concentrated on the highest selling product volume over the period, the Original Sundae as it appears in Table I. This enables the forecasting model to be tested on the main product with the great influence on the store revenue.

TABLE I
TOP SELLING PRODUCTS

Product Name	Total Sold
Original Sundae	14,314
B Sundae	14,265
Mini Ice Cream Coklat	13,229
Aku	12,490

Preprocessing of data was conducted to process the dataset so that it could be used by the time series model. To begin with, all POS files were received monthly and merged into one dataset before being sieved in order to contain only the transactions on the Original Sundae product. Subsequently, it is aggregated according to the sales timestamp into hourly resolution, such that a time series of the number of sales per hour is obtained as presented in Table II.

TABLE II
SAMPLE OF HOURLY AGGREGATED SALES

Timestamp	Total (IDR)
2023-11-12 18:00:00	30,000
2023-11-12 19:00:00	40,000
2023-11-12 20:00:00	20,000
2023-11-12 21:00:00	0
2023-11-12 22:00:00	0

It is also a process that involves sorting by date, missing values checks, and aligning the date data type. Before modeling, a visual examination of the time series was performed analyzing demand trends and Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF). The analysis showed that there was a strong daily seasonality, was characterized by the ACF peak at lag 24 and a strong reliance on the last hour (lag 1), hence explaining the statistical soundness of the forecasting methodology and the forecasting base.

The modeling stage consisted of the development of a Long Short-Term Memory (LSTM), based foretelling framework. MinMaxScaler was used to normalize the data, and 168-hour or one full week input sequences were created in order to capture daily and weekly seasonal trends. The data was further split by time sequence into train (70), validation (15), and testing (15). The architecture of the LSTM model employed is simple consisting of one memory layer of 32 units and one dense output layer. Overfitting was also prevented by early stopping during the training. Three base techniques were used to make the comparison: Naive Forecast, 7-hour Moving Average (MA7), and Daily Seasonal Naive (t-24). The two standard measures of error in time series forecasting, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), were used to evaluate the performance of the model.

This study, as an extension of the error-based evaluation, also involved an operational perspective in terms of simulating the cost of stock decisions. This method turned the predictions of each of the models into hourly stock quantities decisions and these were matched with the real demand. The definition of understock and overstock costs adopted a basic cost method which responds to possible business loss like the lost sales margin against storage costs or wastage. An overview of the total costs during the period when the test was carried out demonstrates the practical advantages of the model in the business setting.

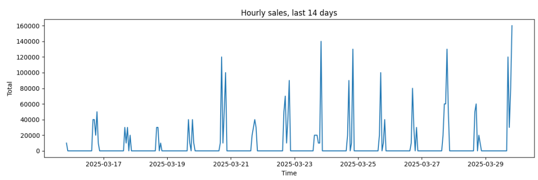

Fig. 1. Hourly sales during the last 14 days.

The following is a figure of the trend of sales of Original Sundae products in the past 14 days of observation, as illustrated by Figure 1. The hours with zero sales reveal the time when the business is not operating or the time when the demand is very low. Nevertheless, there are operating hours where there are high peaks of sales around 100,000 Rupiah an hour. The trend validates the rapid change in demand in a given day and states that the seasonality in a day is high.

The Autocorrelation Function (ACF) graph of hourly sales was plotted in figure 2. It is observed

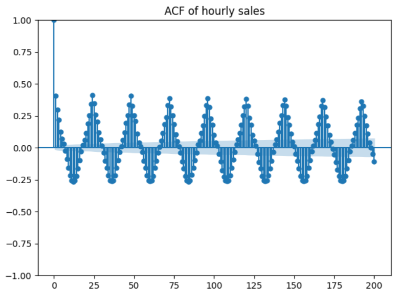

Fig. 2. Autocorrelation Function (ACF) of hourly sales, indicating strong daily seasonality.

that lag 1 is highly correlated with the present value, which shows that the present pattern of demand heavily depends on the short term. Moreover, recurring peaks, at lags of 24, 48, 72, and so on, are observed, which implies a definite daily seasonality, such as demand patterns, which repeat after every 24 hours. The continuous positive and negative fluctuation of correlation reflects the fact that the time series is non-stationary and cannot be only affected by the linear tendency but also by the intricate seasonal changes

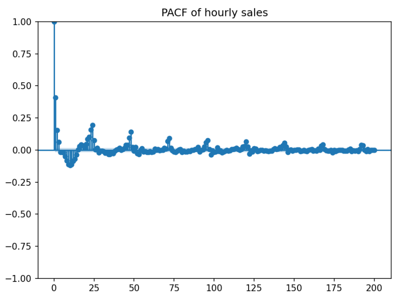

Fig. 3. Partial Autocorrelation Function (PACF) of hourly sales, highlighting short-term dependency significance.

The Partial Autocorrelation Function (PACF) of hourly sales is shown in Figure 3. As can be observed, lag 1 is associated with the current value with a very high degree of partial correlation, which means that there is a very strong direct correlation between consecutive hours. PACF value declines sharply after lag 1 and moves close to zero, which means that the first-order long-term dependence is not that strong

IV. Result

A Long Short-Term Memory (LSTM) was used to predict hourly sales of Original Sundae. The model is a LSTM layer with 32 memory units and a Dense layer which has a single neuron as an output. The Adam optimizer and Mean Squared Error (MSE) loss function were used to train the model. To avoid overfitting, early stopping mechanism was applied with a value of 10 based on validation loss monitoring. The length of the input sequence was made 168 hours (one week) and this was selected according to the findings of ACF and PACF analyses which showed that there was a strong daily seasonal variation. The model was also trained up to 100 epochs at a batch size of 64.

Table III demonstrates the performance of the LSTM model on the test set. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) of the LSTM model were 10,017.062 and 18,022.698 respectively. Such findings suggest that the model has the capacity to reproduce the time trend of hourly sales at a very small error margin although the data is dynamic and has a high proportion of zero values during non-operative periods.

The sales data will cover the time frame between November 12, 2023, and March 29, 2025, and it will have a total of 12,075 hourly data points. The range of sales values is between 0 and 400,000 Rupiah. The median of 0 means that there were no sales in most of the hours and the mean value was 11,854 Rupiah which verifies the high level of fluctuation and seasonality of operating hours. Three baseline techniques were also examined in this study: Naive Forecast, 7-hour Moving Average (MA7) and Daily Seasonal Naive (lag 24). Table III demonstrates a summary of model performance. The LSTM model was found to have the most optimal performance and the reduction in MAE and RMSE was significantly lower than the Naive and MA7 baselines. Even though the Daily Seasonal Naive baseline had an insignificantly lower MAE, its value of RMSE was largely greater than that of the LSTM, which suggests that the model is more likely to make some huge errors when the demand is higher.

TABLE III
MODELS PERFORMANCE COMPARISON

Metode	MAE	RMSE
LSTM	**10.017,062**	**18.022,698**
Naive	10.408,506	23.331,868
MA7 (rata-rata 7 jam)	12.233,592	21.984,527
Daily Seasonal Naive (t-24)	10.005,596	22.441,867

Figure 4 presents the results of the comparison of the forecasting outcomes of the LSTM model and the three baseline methods on the test set in the final 7 days of the observation period. As can be observed the actual sales pattern is sharp peaked at the operating hours and at the non-operating hours the value is zero. Naive and Daily Seasonal Naive models could track the general

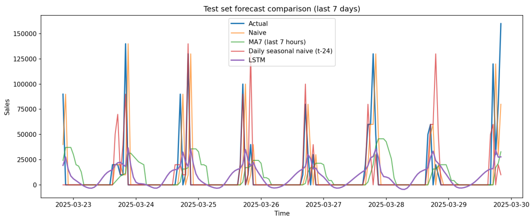

Fig. 4. Last 7 days of models comparison

trend of the peak, although in many instances they were large in terms of error. In the meantime, the MA7 model was slower in responding and tends to flatten out the signal hence overlooks the dynamics of the daily sales spikes. Because of its contrast with the baseline, the LSTM model has always adhered to the dynamic sales trend, especially in the period preceding the daily peak demand. Though discrepancies between the actual values still persisted at the peak, the form of the LSTM movement pattern was closer to the actual demand dynamics. This means that LSTM can learn the temporal nature of the relationships between hours and the intricate daily seasonal aspects, leading to improved general prediction accuracy. Following these findings, it could be concluded that LSTM is the most ideal model when it comes to predicting hourly sales particularly when demand is surging and thus may pose risk of stock-out

V. Business Implication

In order to quantify the operational impact of the forecasting results, this research used a cost-based simulation within the test period. The comparison between the forecast value and actual hourly demand was used to conduct the simulation. When the forecast number is less than the real demand, there will be an understock state leading to loss of sales and a fine of per unit of understock is paid. On the other hand, when the forecast value exceeds the actual demand, an overstock situation arises which may result in waste of material or storage expenses and a penalty of per unit of excess is charged. This simulation was implemented on all the models: Naive, 7-hour Moving Average (MA7), Daily Seasonal Naive (t-24), and the LSTM. The summary of the simulation results was then in the form of total operational costs, and average hourly costs in order to evaluate the financial effect of the models in terms of inventory management. The simulation results are shown in Table IV. The final cost obtained by the LSTM model was 26,948,010 Rupiah with an average cost per hour of 15,080 Rupiah. This is less than the Naive and MA7 models which incurred expenses of IDR 27,980,000 and IDR 32,951,430 respectively. Daily Seasonal Naive model had the least total costs, however, the difference was incredibly low relative to LSTM. It must be mentioned that although the Daily Seasonal Naive model is cheaper, it does not perform as well concerning error when the sales are at a peak which can be expensive in situations where loss of sales might occur.

TABLE IV
Operational Cost Simulation

Model	Cost	Average Cost
Naive	27.980.000	15.657,53
MA7	32.951.430	18.439,52
Seasonal Naive	**26.840.000**	**15.019,59**
LSTM	26.948.010	15.080,03

Based on the results, one can arrive at the conclusion that LSTM model is the most efficient and cost effective in comparison with other baseline methods. This model can be applied in the operation of the store to minimize the risk of stockout at the busiest times, as well as minimize waste. Thus, the application of LSTM-based forecasting can help achieve higher profitability and more data-driven inventory decisions.

VI. Conclusion

This paper shows that Long Short-Term Memory (LSTM) model can offer superior hourly sales prediction results than the control strategy on the Original Sundae product. According to ACF and PACF, daily seasonality, and time-dependent behaviour of hours, there is a solid foundation to the assumption that a deep learning-based approach is necessary to describe intricate demand dynamics. Test set evaluation indicates that LSTM has a reduced error rate compared to Naive and MA7, and is more consistent than Daily Seasonal Naive when sales spikes occur. Moreover, cost-based simulations demonstrate that LSTM has the ability to minimize the possible losses caused by stock-outs and waste caused by overstock, which is why it can add to operational efficiency and improved business profitability. Therefore, it is suggested to make use of the LSTM model as an efficient forecasting tool to assist in the data-driven decision-making in the food and beverage retail industry in terms of inventory. This study has several limitations. First, the analysis focused only on a single product in a single business context, so generalization to other industries requires further research. Second, the LSTM architecture used was simplified to remain realistic for application in

MSMEs. Third, external factors such as promotions, weather, and holidays were not incorporated into the model—this could be a topic for further research.

Author Contributions

Moch Azis Munawar: Conceptualization, Literature review, Data acquisition, Methodology, Writing original draft.
Ivan Diryana Sudirman: Data curation, Formal analysis, Model development, Validation, Visualization, Supervision.

Open Data Statement

The dataset used in this study consists of real transactional records from an active commercial business and contains sensitive operational and customer information. Due to confidentiality and privacy restrictions, the dataset cannot be shared publicly. Only aggregated and anonymized results are presented in this paper.

References

[1] A. Sudirman, O. S. Sinaga, J. Julyanthry, and E. Grace, "Competitive advantage analysis of msmes in the culinary sector: Reviewing the influence of market orientation and green supply chain management," *Journal of Business and Management Inaba*, vol. 3, no. 2, pp. 84–100, 2024. [Online]. Available: https://journals.inaba.ac.id/index.php/jbmi/article/view/366

[2] A. A. Borja-Gonzales, A. B. Perez-Soto, and A. Flores-Perez, "Stockout reduction using forecasting methods, the eoq model and a safety stock in a peruvian sme in the commercial sector," in *Industrial Engineering and Industrial Management*, ser. Communications in Computer and Information Science, S. Sheu, Ed. Cham: Springer, 2024, vol. 2070, pp. 65–75.

[3] H. Sharma, N. D. Kumar, S. Baskar, D. Shukla, D. Singh, S. K. Tiwari, and M. Danish, "Enhancement of sales forecasting and prediction with machine learning methods," *International Research Journal of Computer Science*, vol. 11, no. 11, 2024.

[4] H. Sak, A. W. Senior, and F. Beaufays, "Long short-term memory based recurrent neural network architectures for large vocabulary speech recognition," *CoRR*, vol. abs/1402.1128, 2014. [Online]. Available: https://arxiv.org/abs/1402.1128

[5] M. Krichen and A. Mihoub, "Long short-term memory networks: A comprehensive survey," *AI*, vol. 6, no. 9, 2025. [Online]. Available: https://doi.org/10.3390/ai6090215

[6] A. Ardakani, Z. Ji, and W. J. Gross, "Learning to skip ineffectual recurrent computations in lstms," in *Proceedings of the Design, Automation & Test in Europe Conference & Exhibition (DATE)*, 2019, pp. 1427–1432.

[7] S. Agrahari and S. Singh, "Sales forecasting using machine learning," *International Journal of Scientific Research in Engineering and Management*, vol. 9, no. 5, pp. 1–9, 2025. [Online]. Available: https://ijsrem.com/download/sales-forecasting-using-machine-learning-3/

[8] M. Subhalakshmi, R. Selva Pradeepa, S. Priyadharshini, and S. Maheswari, "Lstm based sales prediction system for supermarket," *International Journal of Advanced Research in Science, Communication and Technology*, vol. 4, no. 3, 2021. [Online]. Available: https://ijarsct.co.in/april3.html

[9] J. R. Muscatello and R. L. Coccari, "Savings with a formal forecasting system: A case study," *The Journal of Business Forecasting: Methods & Systems*, vol. 19, no. 2, pp. 19–21, 2000.

[10] H. Khusna and S. Suhartono, "Pendekatan percentile error bootstrap pada model double seasonal holt-winters, double seasonal arima, dan naïve untuk peramalan beban listrik jangka pendek area jawa timur-bali," *Jurnal Sains dan Seni ITS*, vol. 4, no. 1, pp. D43–D48, 2015.

[11] H. G. Schulze, R. B. Foist, K. Okuda, A. Ivanov, and R. F. B. Turner, "A small-window moving average-based fully automated baseline estimation method for raman spectra," *Applied Spectroscopy*, vol. 66, no. 7, pp. 757–764, 2012.

[12] S. Wang and J. Ma, "A novel ensemble model for load forecasting: Integrating random forest, xgboost, and seasonal naive methods," in *2023 2nd Asian Conference on Frontiers of Power and Energy (ACFPE)*, Chengdu, China, 2023, pp. 114–118.